

7(a)(i)

any **three** from:

- 1 blood glucose concentration rises in both with drink ;
- 2 diabetic, always / initially, has **higher** blood glucose concentration ;
- 3 diabetic has, **higher** peak / **steeper** rise, in blood glucose concentration ;
- 4 diabetic blood glucose concentration, has slower fall / does not return to normal in 3 hours / is still high after 3 hours ;
- 5 compare two data points with x and y figures and units ;

3

Question	Answer	Marks
7(a)(ii)	β cells / pancreas / islets of Langerhans / hypothalamus ;	1
7(a)(iii)	<u>negative feedback</u> ;	1
7(b)(i)	A – <u>glucose oxidase</u> ; B – (hydrogen) <u>peroxidase</u> ;	2
7(b)(ii)	<i>biosensor</i> – rapid / immediate / 2 seconds, reading or (more), accurate / precise / quantitative / numerical / sensitive or shows current blood glucose / re-usable ; <i>dipstick</i> – non-invasive / painless / less risk of infection / easy to use / cheap ;	2
7(c)	any five from: 1 (insulin) binds to receptors on, liver / muscle / adipose (cells / tissue / organ) ; 2 GLUT(4) added to membrane of, muscle / adipose, cells or (GLUT 4) increases , muscle / adipose, cell, permeability (to glucose) ; 3 glucokinase phosphorylates glucose in liver cells or increased (rate of) <u>diffusion</u> of glucose into liver cells ; 4 more / faster, respiration of glucose ; 5 activates glycogen synthase / causes glycogenesis / causes glycogen synthesis ; 6 causes, lipid / triglyceride / fatty acid, synthesis or decreases, glycogenolysis / lipolysis / gluconeogenesis ; 7 <u>blood glucose concentration</u> decreases ;	5